

University of Pennsylvania, College of Liberal and Professional Studies

Syllabus for General Chemistry 101 – Summer 2016

Textbook: Zumdahl – Chemical Principles (7th Edition)

Introductory Material

Units of measurement, physical properties of Atoms and Molecules

Chapter 2 - Atoms, Molecules & Ions - Review

Chapter 20 - The Nucleus - extract parts relating to isotopes to support discussion of average atomic masses in Chapters 2 & 3.

Chapter 3 - Stoichiometry – detailed coverage of Section 1 - rest review.

Chapter 5 – Gases (except the parts on kinetic theory (see Chem 102))

Chapter 13 – Bonding: General Concepts

Chapter 4 - Types of Chemical Reactions and Solution Stoichiometry –

Mostly review - detailed coverage of Sections 10-12.

Chapter 12 - Quantum Mechanics and Atomic Theory

Chapter 14 - Covalent Bonding: Orbitals, Molecular Spectroscopy

Chapter 21 - Organic and Biochemical Molecules - extracts used to support discussion in Chapters 13 & 14.

Chapter 16 - Liquids & Solids - all except for Sections 10 & 11.

A section will be added on liquid crystals if time permits.

Comments:

- This course, although accelerated and given over 5.5 weeks, is in every other way equivalent to a regular semester course at Penn and carries full credit. It is taught by a regular Penn faculty member who also teaches in the regular sequence.
- The course is supported by recitations given by Penn Chemistry Graduate students.
- There are two 1-hour mid-term exams, plus a two-hour final exam.
- The course is run concurrently with the 053 laboratory, which will involve most students from 101. However, the lab is administered separately and requires separate registration. Some lab experiments are designed to complement lecture material.

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